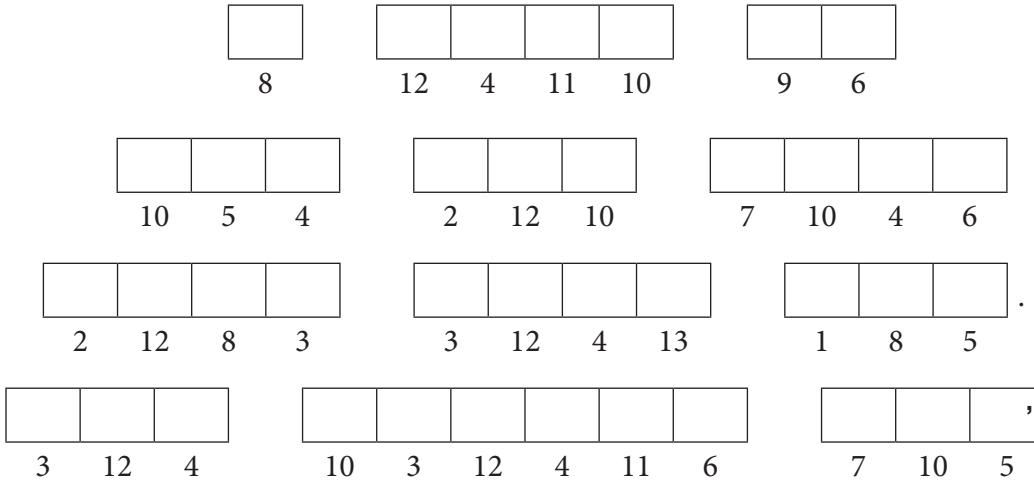
WORKSHEET

Chapter 7: Logarithms – Solving equations 1



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Questions

Solve the following equations. (Remember, log a is possible only if $a > 0$.)

1 $4^{5x-2} - 64 = 0$

2 $3^{x-5} + 3 = 30$

3 $3^{2x} + 81 = 2 \times 3^{x+2}$

4 $8^{x-3} = 4^{x-1}$

5 $\log_3(x - 9) = 4$

6 $\log_2(3x + 20) = 5$

7 $\log_2(x - 1) + 1 = \log_2(x + 7)$

8 $\log(3x - 1) = 5 \log(2)$

9 $\log_5(x - 1) - \log_5(x + 4) + \log_5(2) = 0$

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10 $2^{2x} + 2 = 2^{x+1} + 2^x$

11 $3^{2x} - 3^{x+2} = 3^{x+1} - 27$

12 $[\log_2(x-3)]^2 + 32 = \log_2(x-3)^{12}$

13 $2^{2x} + 144 \times 2^{x+1} = 8192$

Solutions

A $x = 11$

C $x = 1$

D $x = 9$

E $x = 7$

H $x = 5$ or $x = 6$

I $x = 6$

N $x = 3$

O $x = 0$ or $x = 1$

R $x = 1$ or $x = 2$

S $x = 4$

T $x = 2$

W $x = 8$

Y $x = 5$ or $x = 8$

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